LOAD BALANCER

Importance of load balancer:

1️⃣ Distributes incoming traffic evenly across multiple servers to prevent overload.
2️⃣ Ensures high availability by routing traffic only to healthy instances.
3️⃣ Increases fault tolerance by automatically handling instance or AZ failures.
4️⃣ Improves application performance and reduces latency.
5️⃣ Works with Auto Scaling to handle traffic spikes smoothly.
6️⃣ Enhances security with SSL termination, WAF, and DDoS protection.
7️⃣ Provides centralized monitoring and logging for better visibility and troubleshooting.

Tools going to be used for the task:

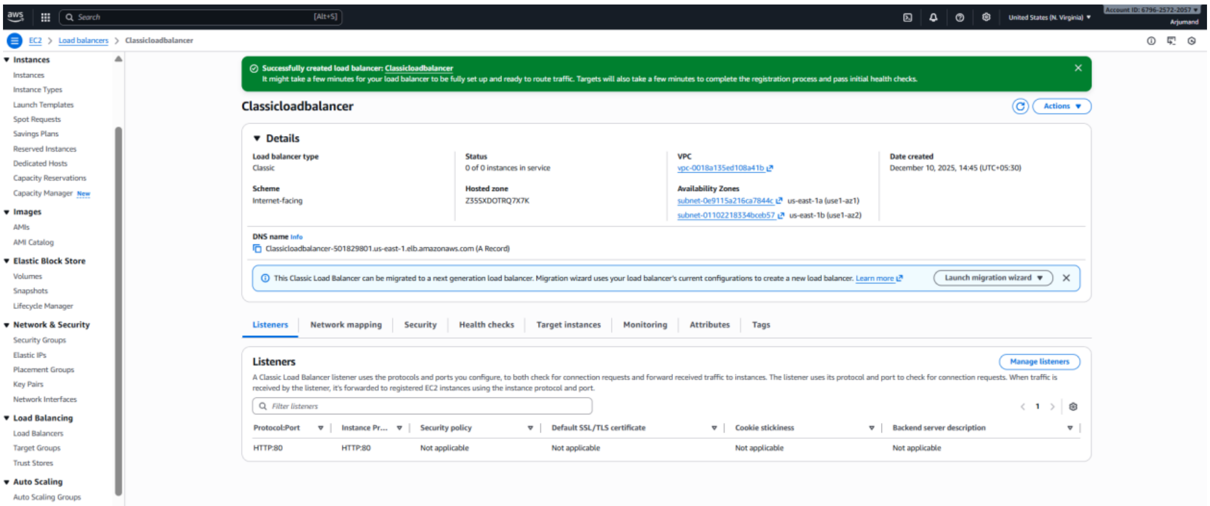
1. **EC2** → to launch instances for all load balancers.
2. **VPC (subnets + security groups)** → networking setup for CLB, ALB, NLB.
3. **Target Groups** → required for ALB and NLB routing.
4. **ACM (AWS Certificate Manager)** → to generate and attach SSL certificate to ALB.
5. **Route 53** → to map ALB DNS using an Alias record.
6. **S3 Bucket** → to store Application Load Balancer access logs.

1. Configure Classic Load balancer.

Steps:

- **Go to EC2 → Load Balancers → Create Load Balancer → Classic Load Balancer.**
- **Choose Internet-facing, select public subnets, and keep listener HTTP:80.**
- **Configure security groups to allow inbound HTTP (port 80).**
- **Add health check using HTTP on path / and default thresholds.**
- **Register your EC2 instances, save, and verify they show InService.**

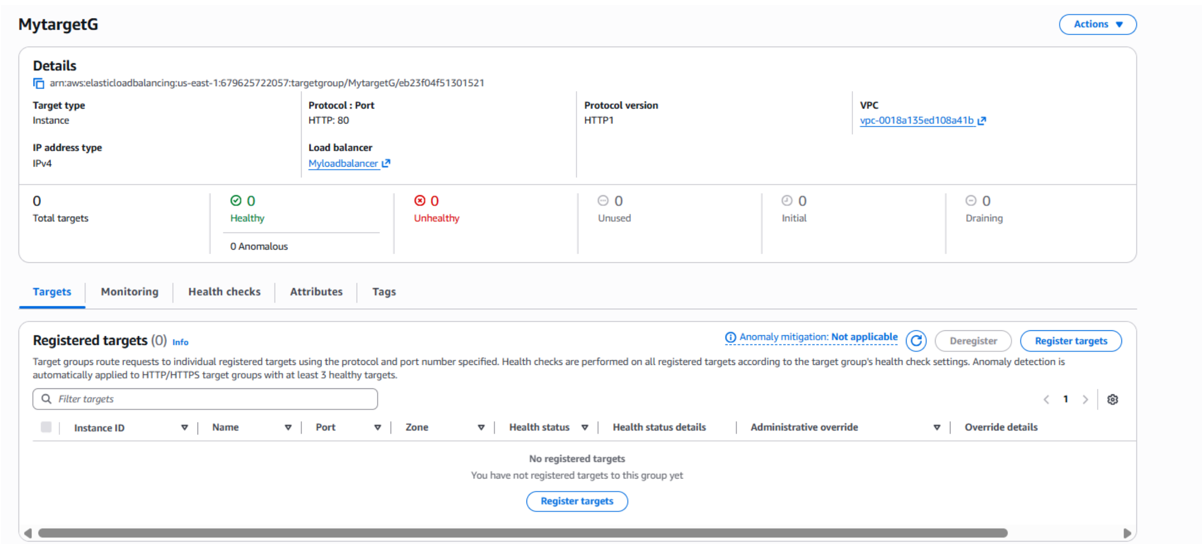
- I have configures classic load balancer with two public ids in different zones



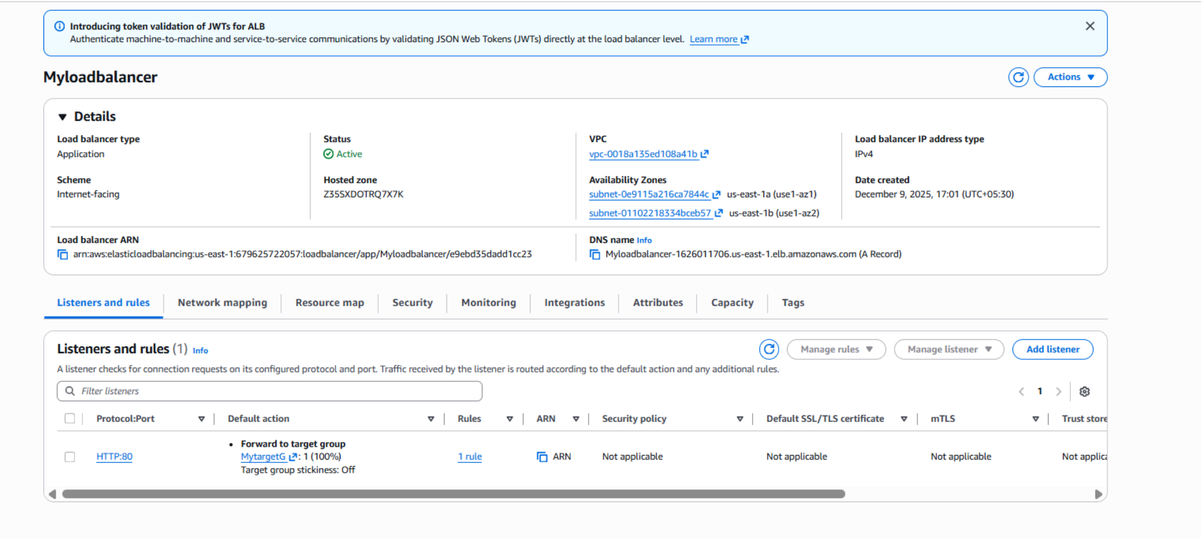
2. Configure Application Load balancer.

Steps:

- Go to **EC2 → Load Balancers → Create Load Balancer → Application Load Balancer**.
- Select **Internet-facing**, choose **IPv4**, and pick **at least two public subnets**.
- Create/choose a **Security Group** that allows inbound HTTP (80).
- Create a **Target Group** (Instance type → HTTP:80) and **register your EC2 instances**.
- In the ALB Listener section, add listener **HTTP:80 → Forward to Target Group**.
- Create the ALB and verify that **Targets show “healthy”** and ALB DNS loads your website.



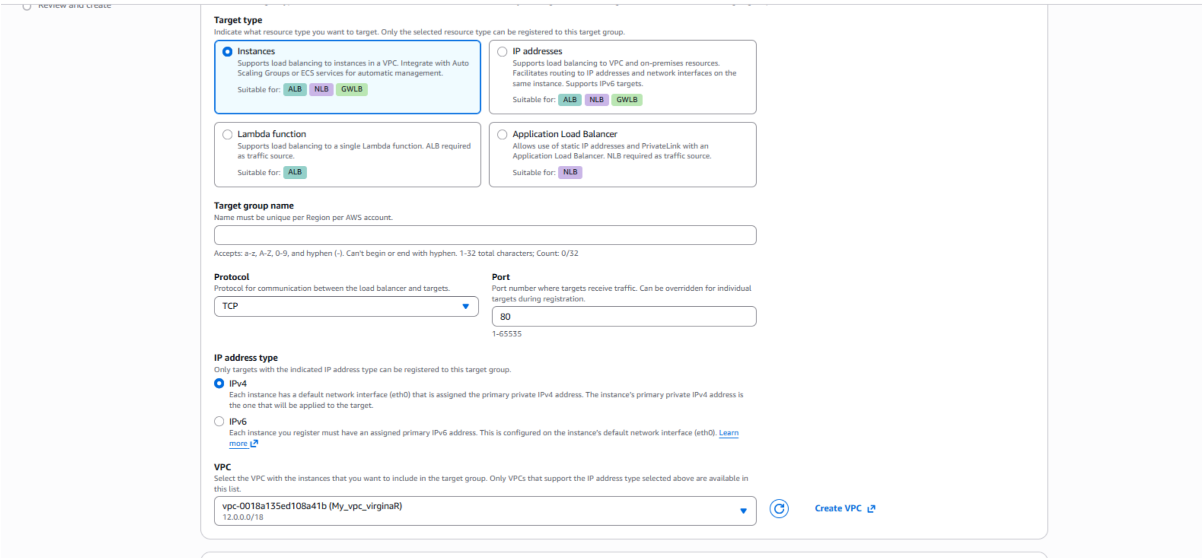
- Created ALB using two public subnets and one target group and security group

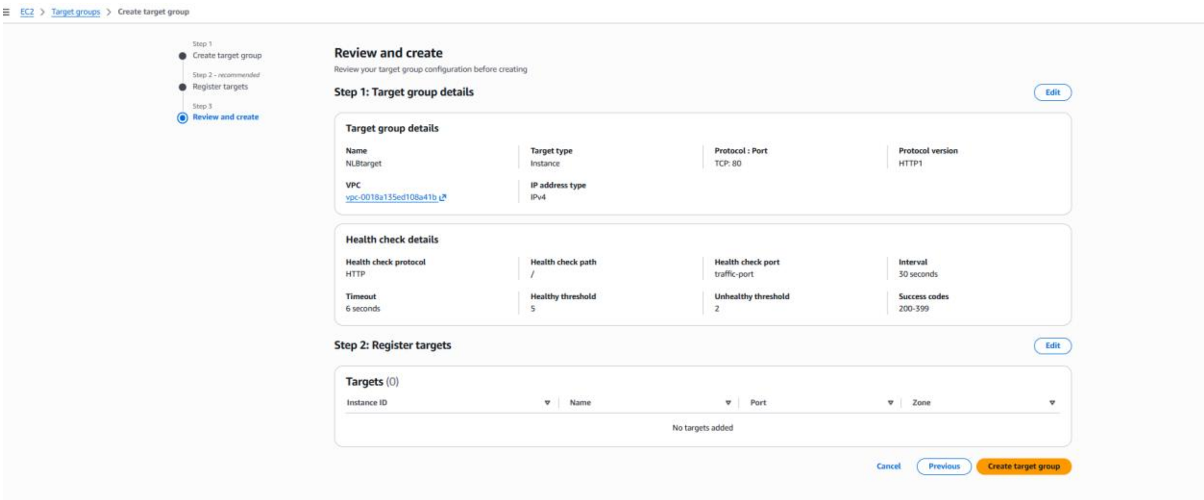


3. Configure Network Load balancer.

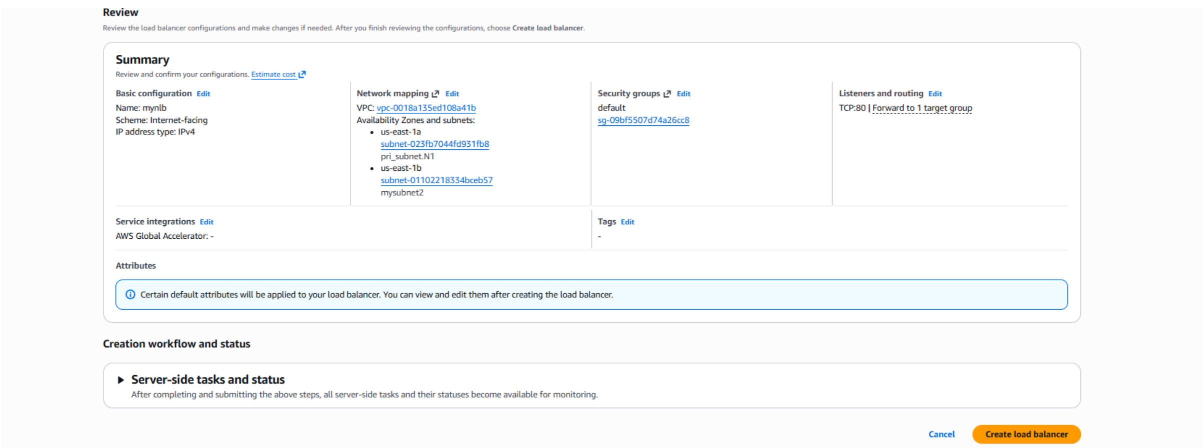
Steps:

- ➢ **Go to EC2 → Load Balancers → Create Load Balancer → Network Load Balancer.**
- ➢ **Enter a name, choose Internet-facing, and keep TCP:80 as the listener.**
- ➢ **Select two public subnets across different Availability Zones.**
- ➢ **Create a new Target Group with Target Type = Instance and Protocol = TCP:80.**
- ➢ **Register your EC2 instances as targets and save the target group.**
- ➢ **Attach the target group to the NLB and click Create Load Balancer.**
- ➢ **Test using the NLB DNS name to confirm traffic reaches your backend EC2 instances.**

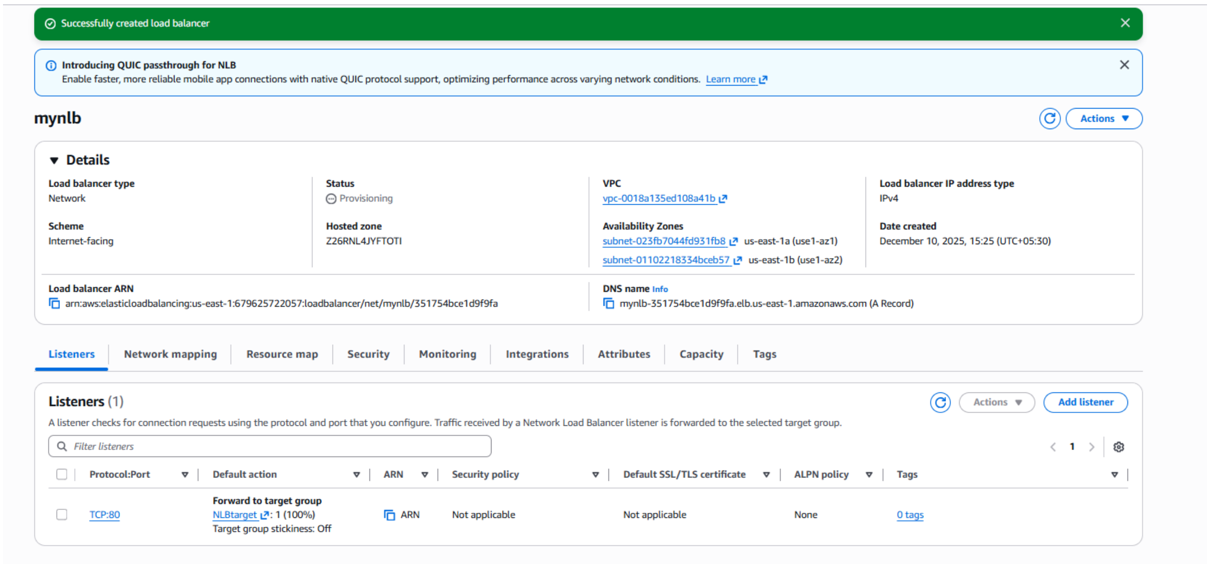




Created a new target group for NLB



NLB created by new target group and two subnets

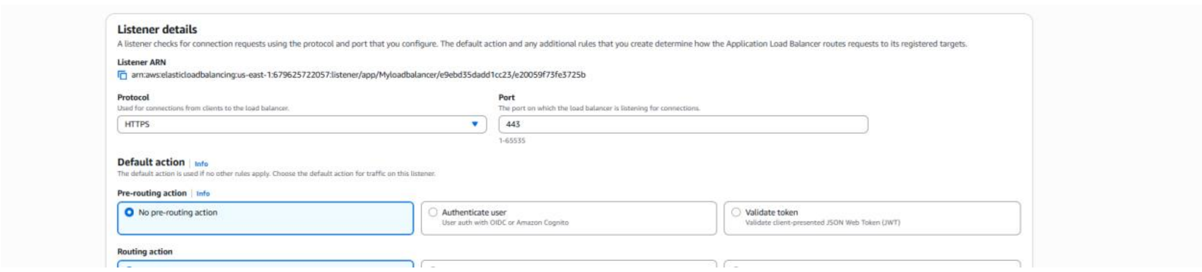


4. Attach SSL for application load balancer.

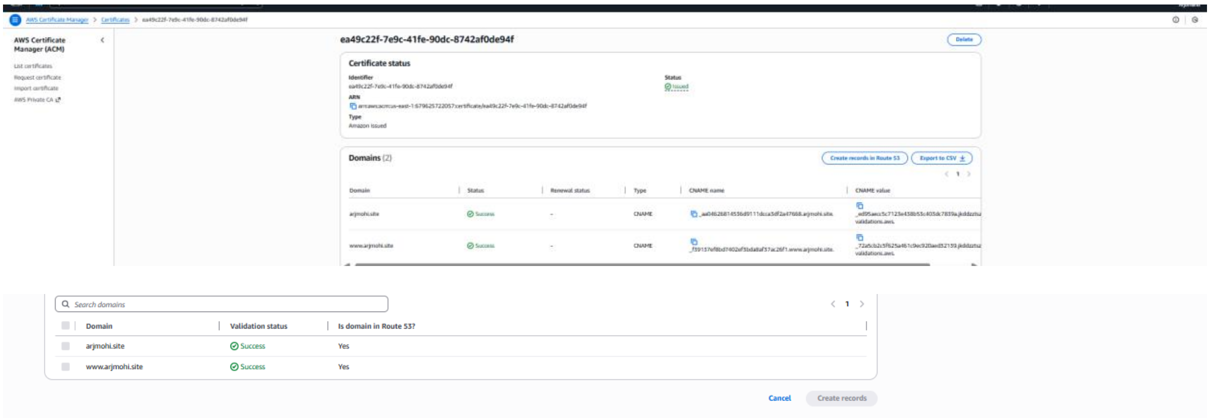
Steps:

- ➢ **Go to EC2 → Load Balancers → Select your ALB.**
- ➢
- ➢ **Open the Listeners tab and click Edit on the HTTPS (443) listener.**
- ➢
- ➢ **Under Secure listener settings, choose a Security policy (recommended TLS13/TLS12).**
- ➢
- ➢ **Set Certificate source = From ACM.**
- ➢
- ➢ **Select your issued certificate (example: arjmohi.site).**
- ➢
- ➢ **Choose the target group you want to forward HTTPS traffic to.**
- ➢
- ➢ **Click Save changes to enable SSL on the ALB.**

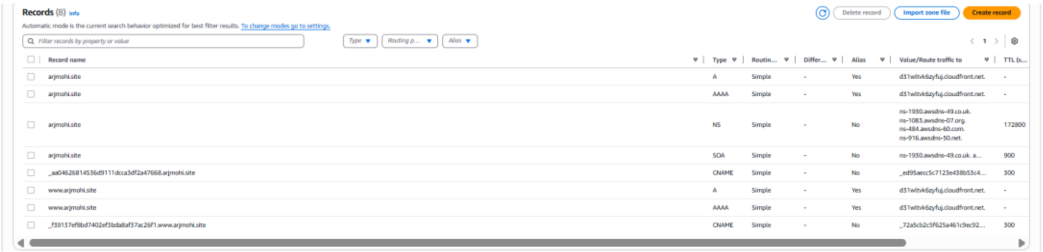
- Created listener details http to https as we are attaching ssl certificate



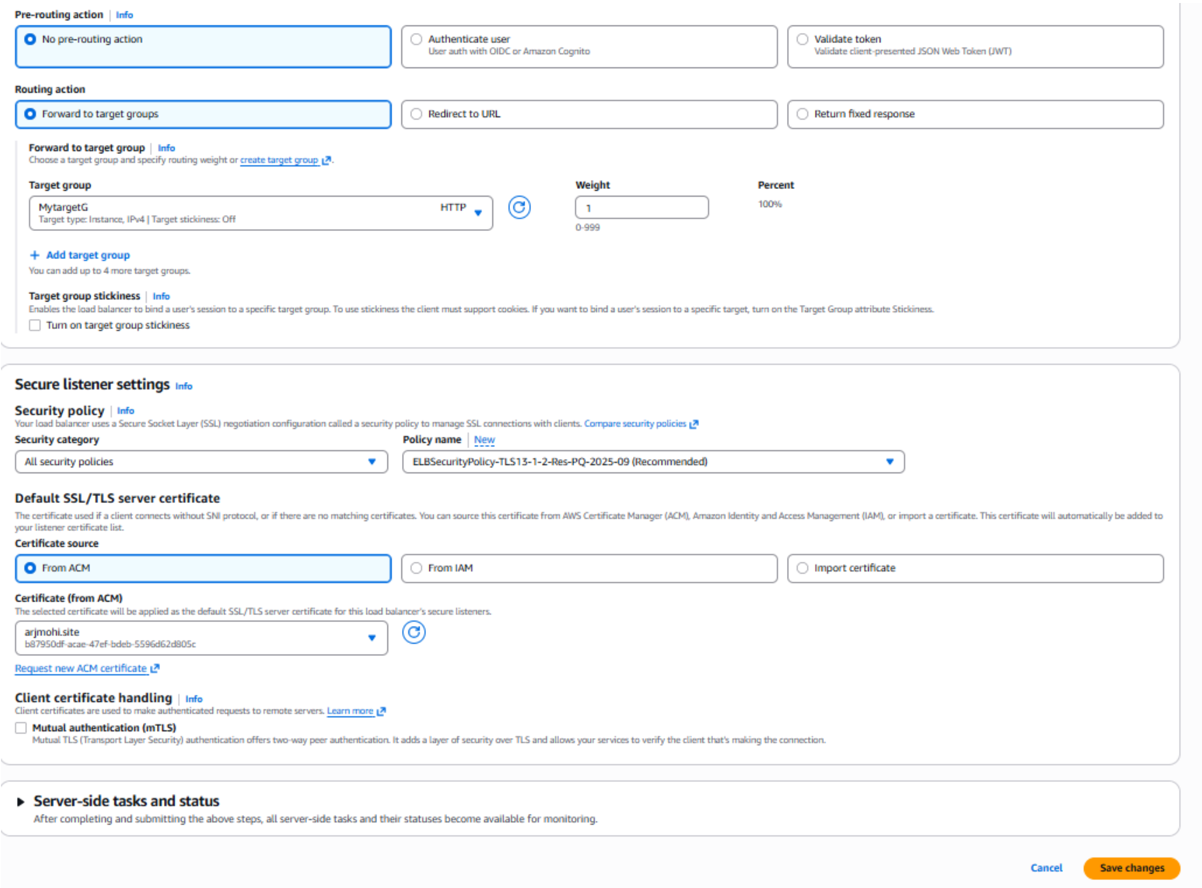
- We can see the records after making certificate with my domain

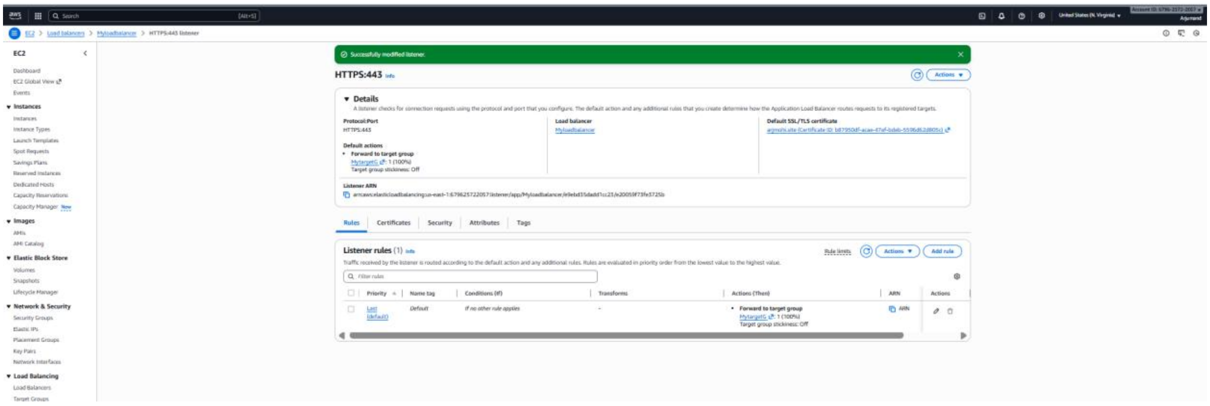


- R53 records as we can see related to my domain



- Added ssl to my APLB and added certificate source



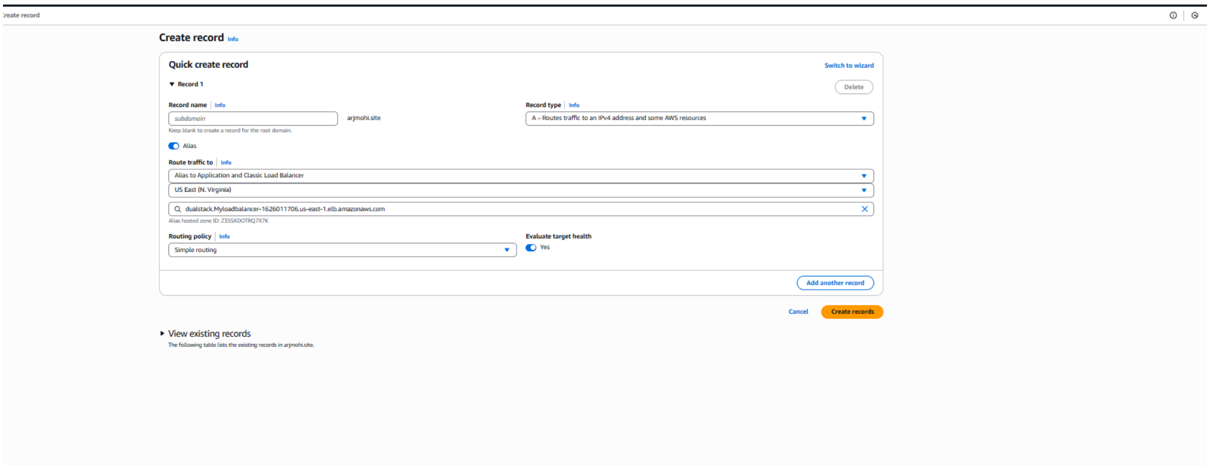


5. Map Application load balancer to R53.

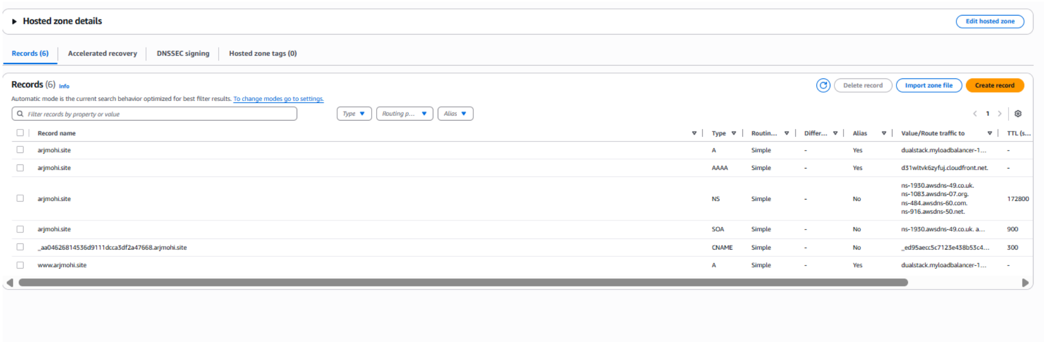
Steps:

- ➢ Open Route53 hosted zone for your domain.
- ➢ Create an **A record** with Alias = **Yes**.
- ➢ Select “Alias to Application Load Balancer.”
- ➢ Choose your ALB from the drop-down list (dualstack).
- ➢ Save the record and verify that the domain resolves to ALB.

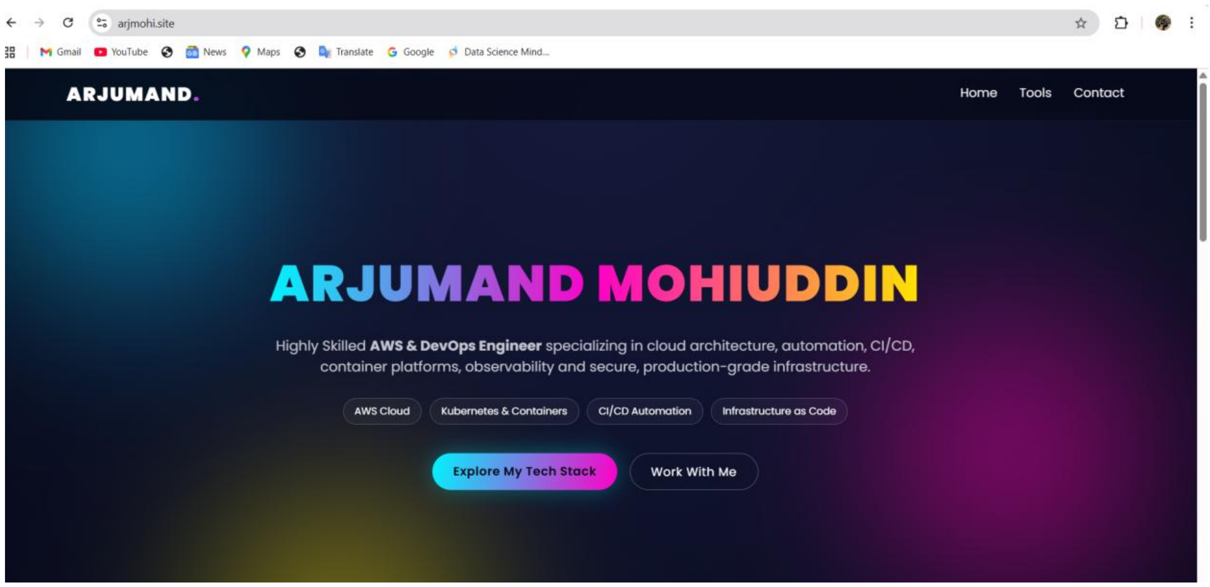
- Mapped application load balancer to route 53



- New records are not added old records are deleted now records redirecting my load balancer



- This is my website is deployed after the new records

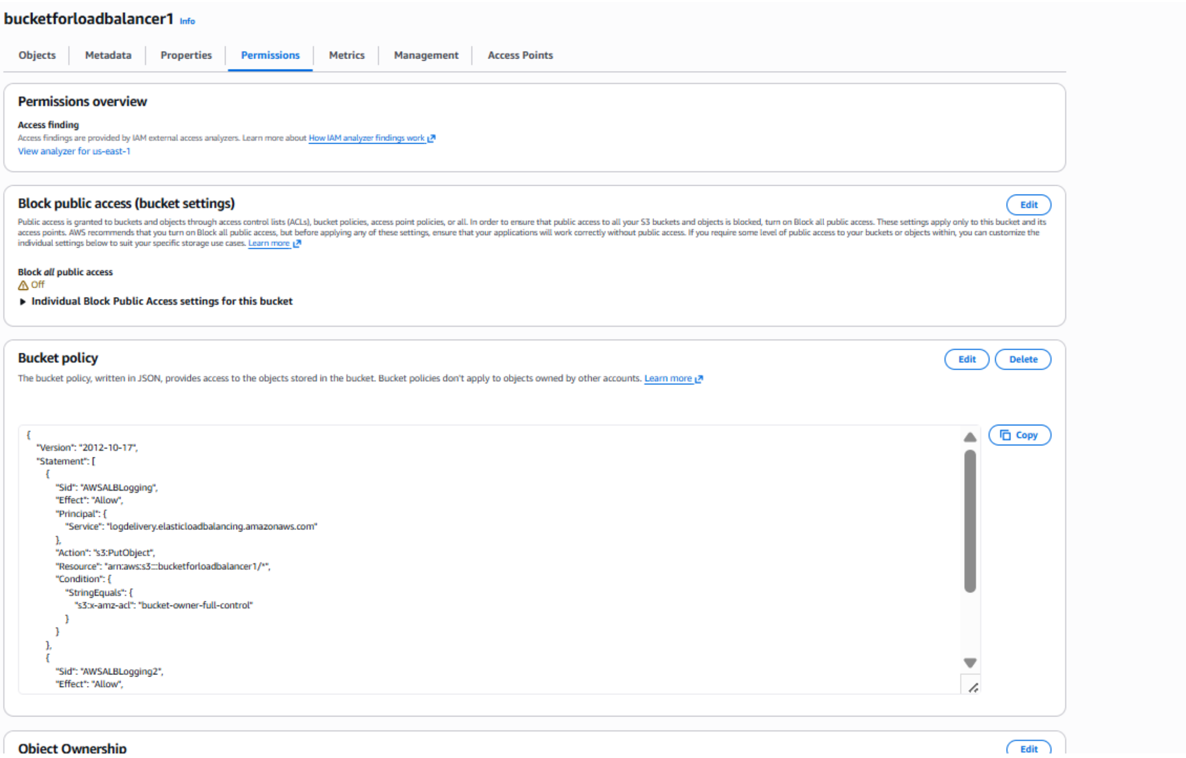


6. Push the application load balancer logs to S3.

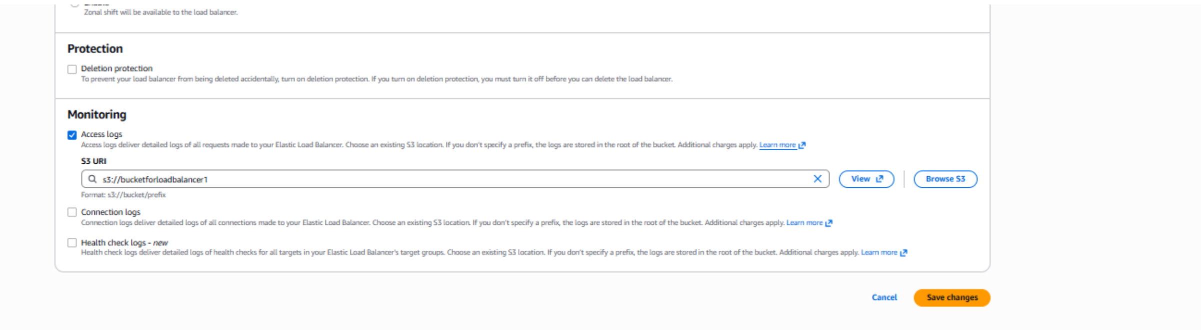
Steps

- **Created an S3 bucket in the same region as the ALB to store access logs.**
- **Added a bucket policy allowing logdelivery.elb.amazonaws.com to write logs.**
- **Opened ALB in EC2 → Load Balancers → Attributes tab.**
- **Enabled "Access Logging" and selected the S3 bucket.**
- **Added optional prefix for log organization.**
- **Saved configuration to start log delivery.**
- **Verified log files in S3 after 5–10 minutes.**

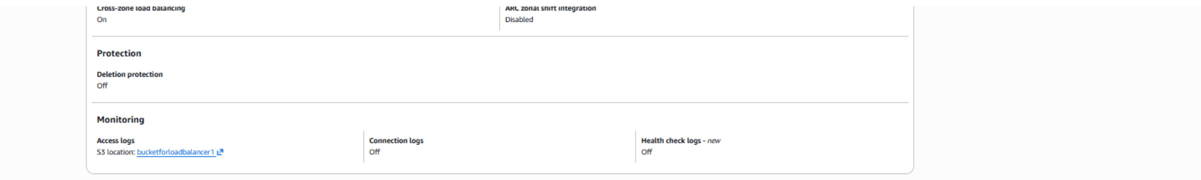
- Created bucket then added bucket policy to receive load balancer logs



- Added bucket in load balancer in attributes tabs



- Bucket is added in load balancer successfully



- We can check load balancer logs are receiving in the s3 bucket